Submission Summary

Conference Name

2nd International Conference on Intelligent Systems using Semiconductors for Robotics and IoT

Track Name

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Paper Title

Evaluating the Robustness of Lightweight Machine Learning Models to Label Noise

Abstract

Label noise is a common problem in real-world datasets and can compromise the reliability of machine learning models. In this work, we study the robustness of six widely used models - Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbours (KNN), and LightGBM, with respect to two types of label noise: symmetric and asymmetric. We conducted experiments on two benchmark datasets: Breast Cancer Wisconsin and Adult Income, to explore whether robustness patterns are consistent under different data settings.

The training data are corrupted at varying levels(0%, 10%, 20% and 30%) and the evaluation is performed on clean test data using Accuracy and F1-score. Results show that performance degrades more due to symmetric noise than asymmetric noise across all models. The decision Tree model is the most sensitive to label corruption whereas, SVM consistently shows stable robustness. F1 score proves to be more sensitive to noise than accuracy, especially for data with class imbalance.

These results highlight the need for robust models to noisy labels and the need for evaluation metrics that capture the true effect of noise in real-world applications where the quality of data is unknown.

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Submission Files

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